# Cybersecurity Home Lab

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# Introduction

This report documents the design and implementation of a personal cybersecurity home lab intended to emulate a small enterprise environment and showcase practical defensive skills. The lab is built entirely on virtual infrastructure and focuses on end-to-end security operations: network segmentation, identity services, monitoring, detection and vulnerability management.

## Scope and Objectives

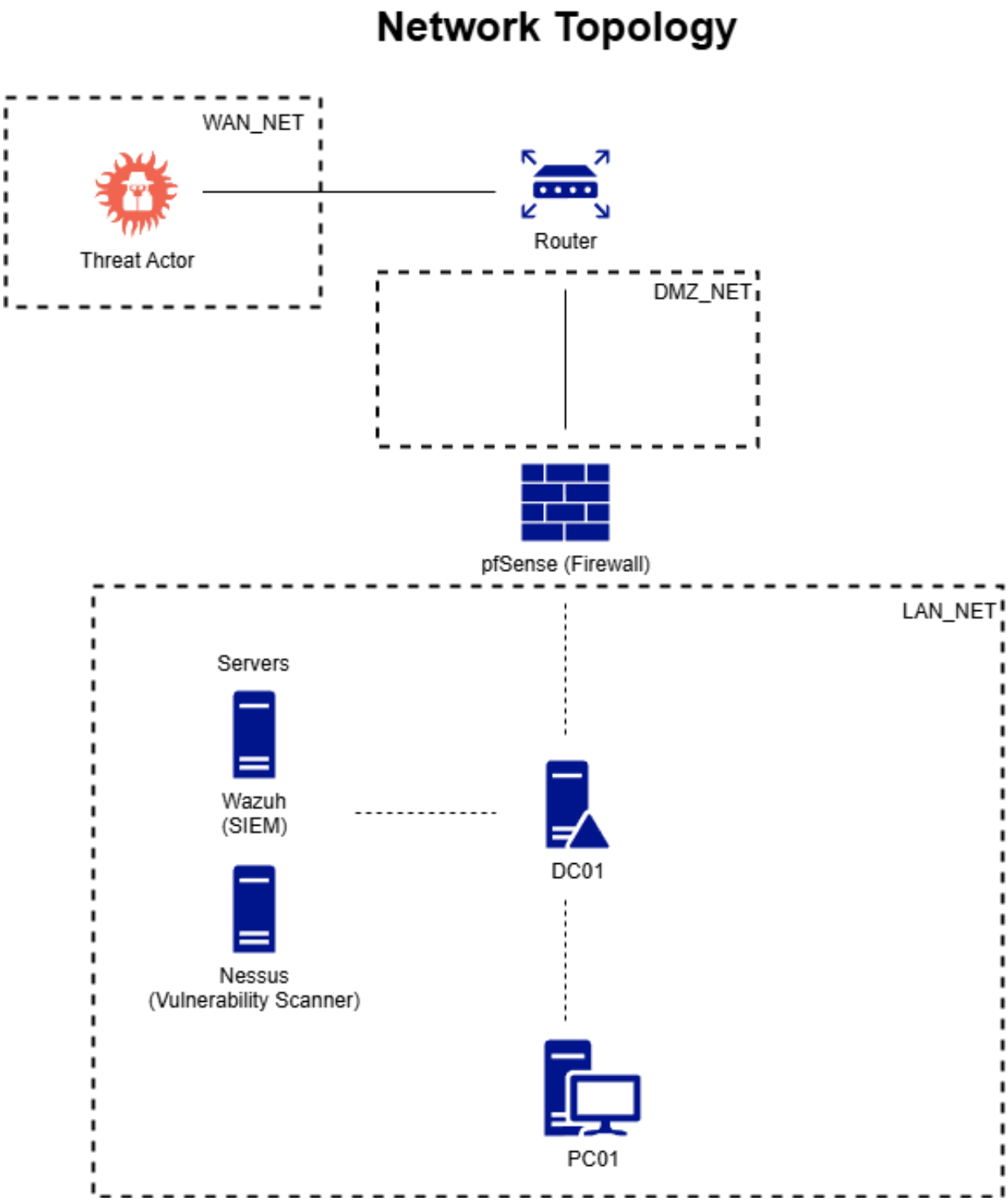
This home lab is scoped to emulate a small, on-premises enterprise network with a single perimeter and one internal LAN, focusing on core defensive capabilities rather than large-scale or high-availability design.

The key objectives are:

1. Design and implement a segmented virtual network (WAN, DMZ, LAN)
2. Deploy and configure pfSense as the primary firewall and internal gateway, including routing between security zones and enforcing security policies
3. Deploy and configure an upstream edge router that handles NAT is the gateway from WAN to the DMZ
4. Set up an Active Directory domain and integrate at least one client workstation to practice identity and endpoint management
5. Install and configure Wazuh (indexer, server, and dashboard) to collect logs from key hosts, generate security alerts, and support basic incident triage.
6. Deploy Nessus for vulnerability scanning of internal assets and use the results to identify misconfigurations and prioritize remediation
7. Simulate common attack scenarios from the external “Threat Actor” system and verify that firewall rules, Wazuh detections, and Nessus findings accurately reflect those activities
8. Produce structured documentation of the lab design, configuration steps, security findings, and lessons learned.

# Lab Design & Architecture

## Network Topology



# Segment Rationale

## WAN_NET

This segment represents the untrusted internet and is where the threat actor will reside. This zone is included in the lab as we want to simulate attacks that originate from outside the trusted network as well as configure proper safeguards and rules for handling traffic from the WAN.

This segment uses a public or simulated-public IP space. Particularly address block TEST-NET-3 with IP address space 203.0.113.0 /24 as detailed in RFC 5737. This results in a NAT being required at the router boundary. Traffic from this segment is untrusted by default.

## DMZ_NET

This segment represents the demilitarized zone of the network which acts as the buffer between the untrusted WAN and the trusted LAN. This segment was created to enforce a two-boundary principle where an attacker must defeat two devices to reach the LAN. This is because traffic must pass through the router inbound from WAN_NET and through pfSense outbound to LAN_NET. In real-world networks, publicly accessible services (web servers, mail servers, DNS) would live here.

This segment uses class C private addressing, 192.168.10.0 /24, which is not directly routable from WAN and is only reachable via NAT on the edge router.

## LAN_NET

This segment represents the trusted zone of the network. The pfSense firewall that acts as the gateway between the DMZ_NET and the LAN_NET enforces a default-deny policy on traffic entering from the DMZ. Only explicitly defined traffic flows will be allowed.

This segment uses class C private addressing, 192.168.20.0 /24, and has no direct exposure to WAN_NET.

## Asset Inventory

| VM Name | Role | OS | vCPU | RAM | Disk | NICs (Network / IP) |
|---|---|---|---|---|---|---|
| EDGE-RTR01 | “ISP” / Core Router | Ubuntu Server-25.10 | 1 | 512MB | 10GB | |
| PFSENSE-FW01 | Firewall / Gateway | | | | | |
| DC01 | Domain Controller | | | | | |
| PC01 | Client Workstation | | | | | |
| WAZUH-SIEM01 | SIEM Server | | | | | |
| NESSUS-SCAN01 | Vulnerability Scanner | Tenable-Core-OL8-Nessus-20241009 | | | | |
| ATTACKER 01 | Threat Actor Box | Kali Linux | | | | |

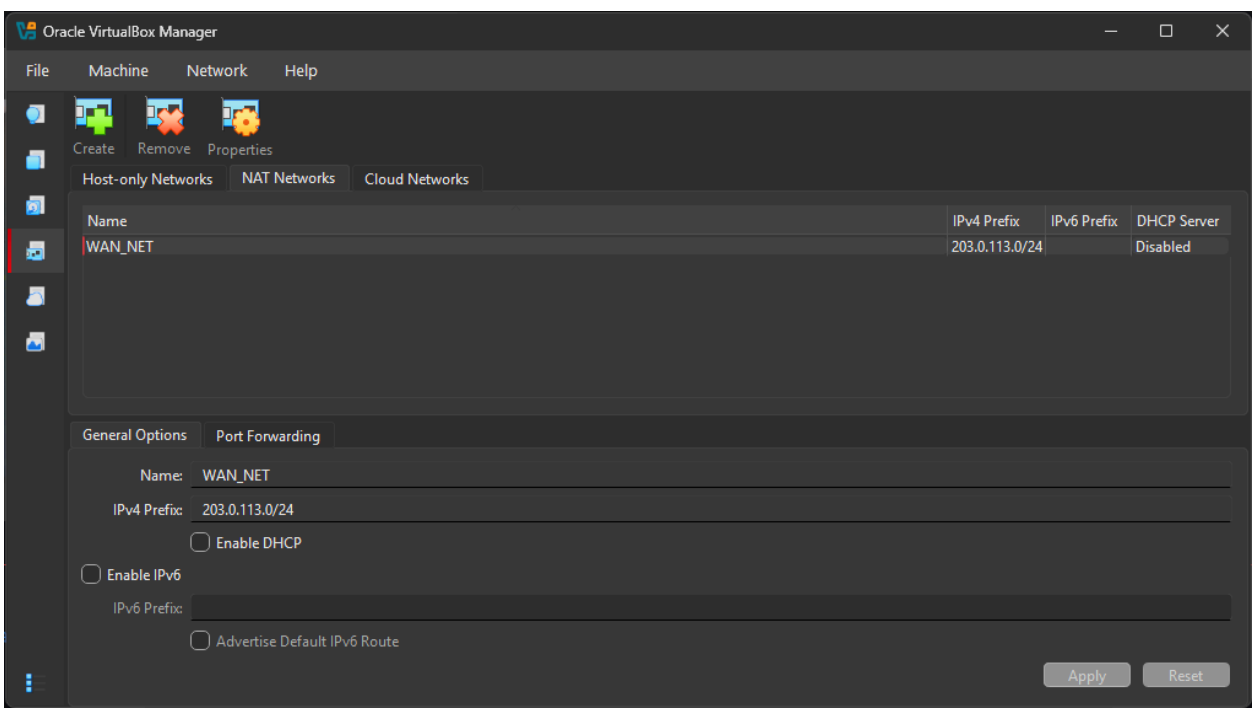
# Environment Setup

## Virtual Box Configuration

### Network Configuration

| Network Segment | VirtualBox Type | IP Address Space | Why |
|---|---|---|---|
| WAN_NET | NAT Network | 203.0.113.0/24 | Simulates internet, allows outbound access |
| DMZ_NET | Internal Network | 192.168.10.0/24 | Isolated segment, no internet by default |
| LAN_NET | Internal Network | 192.168.20.0/24 | Fully isolated internal segment |

### WAN_NET Creation




DHCP in this case is disabled as I will be assigning IP addresses manually.

## VM Account Credentials

| VM Name | User Name | Password |
|---|---|---|
| EDGE-RTR01 | router-vm | P@ssw0rd123 |

## EDGE-RTR01 Configuration

1. Set up the virtual box vm with 512mb ram, 10gb disk storage
2. Complete installation
3. Unmounted the ubuntu-server-iso so we boot up from the disk
4. On first boot, sudo apt update && sudo apt upgrade
5. Shut down the system and changed the network adaptors, configuring the NICs
6. Set adaptor 1 to be connected to the NAT network, WAN_NET
7. Set adaptor 2 to be connected to the internal network, DMZ_NET
8. Sudo su -
9. Cd /etc/netplan
10. Mv 00-installer-config.yaml 00-installer-config.yaml.bak , change the initial netplan installation to just be a backup file
11. Create new netplan config, 01-EDGE-RTR01-config.yaml
12. Chmod 600 ./*.yaml , set permissions for the yaml files properly otherwise the permissions will be too open
13. Netplan apply
14. Ip addr show , shows the applied netplan
15. Ping 8.8.8.8 results in host unreachable
16. Initially netplan address for enp0s3, NIC WAN facing, set the default route via 203.0.113.2 which is incorrect as VirtualBox reserves .1 for NAT Network gateway , different from just the standard NAT on virtual box which reserves .2 as the gateway
17. Was diagnosed by ip neigh show, a command to show the ARP cache
18. When default route via 203.0.113.2 was set, accessing the ARP cache shows us a FAILED
19. Meaning that when I tried to ping 8.8.8.8 initially, it sent out an ARP broadcast for 203.0.113.2
20. This host doesn't exist on the network and as a result no reply, thereby no ethernet frame can be built so the ping cannot be sent out
21. Meaning that the VirtualBox's gateway is not responding on the subnet
22. Meaning that either the WAN_NET was not properly configured as 203.0.113.0/24 or that there is no gateway at 203.0.113.2 at all.
23. Fixing this was simple as we just had to change the default gateway route to instead be 203.0.113.1 where the VirtualBox NAT Network default gateway actually was

24. We then need to enable ip forwarding, otherwise the router just drops the packets instead of forwarding them between interfaces
25. We will do this by adding a .conf file into /etc/sysctl.d
26. We add 99-ip-forward.conf and write net.ipv4.ip_forward=1 into the file
27. This ensures ip forwarding is enabled and persists through reboots
28. Next we need to allow DMZ devices to reach the internet through the router VM
29. We do this by sudo iptables -t nat -A POSTROUTING -s 192.168.10.0/24 -o enp0s3 -j MASQUERADE
30. This commands means
    a. -t nat : work on the NAT table
    b. -A POSTROUTING : append rule to the POSTROUTING chain – runs after routing decision is made, just before packet leaves
    c. -s 192.168.10.0/24 : Match packets sourced from internal LAN
    d. -o enp0s3: Packets that are exiting through WAN interface
    e. -j MASQUERADE: Rewrite the source IP to instead be the router's WAN IP
31. What happens to a packet
    a. DMZ device (eg. 192.168.10.4) sends out a packet
    b. Router VM receives it on enp0s8 (DMZ facing interface)
    c. The routing decision is to send this packet out through enp0s3
    d. POSTROUTING hook then fires and performs MASQUERADE
    e. Rewrites src ip of packet to instead be ip of enp0s3
    f. Packet leaves to WAN_NET with enp0s3 ip
    g. Reply come backs with dest ip as enp0s3 ip
    h. Router translate it back (Using conntrack) to DMZ device ip, forwards the reply back to the DMZ device
32. Router setup is done

# References

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